

# MilkShield – Video Script

Smart India Hackathon 2026 · Problem Statement 26110 · Team INVICTUS (ID 159400)

**≈ 2 min**

total length

**9 scenes**

6 slides + 3 simulation  
shots

**≈ 295 words**

calm pace, ~150  
words/min

**1280 × 720**

all clips, drop straight into  
editor

## THE THREE THINGS THE JUDGES MUST REMEMBER

**1**

### Ice inside the milk

Cold goes straight into the milk, and the foam keeps the heat out. No power, no compressor.

**2**

### Sized by simulation

Three models: below 8 °C in 75 min, held for 14 h on a 35 °C day.

**3**

### Modular and cheap

Carry 2–8 cartridges to suit the load. Prototype costs about ₹6–13 k.

## RUNNING ORDER

| Time | On screen                         | Purpose                                              |
|------|-----------------------------------|------------------------------------------------------|
| 0:00 | Slide 1 – Title                   | Who we are, what problem we solve                    |
| 0:12 | Slide 2 – Proposed solution       | The core idea in one picture                         |
| 0:30 | Slide 3 – Technical approach      | How a farmer uses it; cartridge count                |
| 0:45 | Clip – thermal model animation    | Proof 1: how cold, how fast, how long                |
| 1:00 | Image – wall heat map             | Proof 2: insulation works; weak spot found and fixed |
| 1:12 | Clip – CFD of the milk            | Proof 3: the whole can cools evenly                  |
| 1:30 | Slide 4 – Feasibility             | Practical and affordable                             |
| 1:42 | Slides 5 → 6 – Impact, comparison | Who benefits; why we beat existing options           |
| 1:55 | Slide 6 – Simulation study box    | Next step, and close                                 |

Files: deck in **1\_Presentation**, clips in **2\_Video**, extra charts in **3\_Simulation\_Results** (folder ~/Downloads/MilkShield\_SIH2026).

# Scene by scene: what to show and what to say

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TITLE PAGE

- Problem Statement ID – 26110
- Problem Statement Title- Development of a Low-Cost Light-weight Milk Chilling Can for Small-Scale Dairy Farmers
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Hardware
- Team ID- 159400
- Team Name- INVICTUS

Slide 1 – Title page

0:00 – 0:12

## Introduce the team and the problem

### SHOW

- Hold the title slide full-screen.
- Cursor or slow zoom on "Team Name – INVICTUS".

### SAY

"Hello, we are Team INVICTUS, solving problem statement 26110: a low-cost, lightweight milk chilling can. Milk leaves the cow at about 35 °C and travels for hours in plain metal cans."

**Why it works:** it names the problem in one breath, so the judges know exactly what to judge you on.

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**MILKSHIELD**  
Passive Cooling Milk Chilling Can, 30–40 L

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**Proposed Solution**

**WHAT IT IS**

- Double-wall PUF-insulated can with a food-grade HDPE chamber
- Reusable ice/PCM cartridges sit in the milk – no compressor, no mains power

**HOW IT WORKS**

Passive cooling stores cooling energy in reusable PCM/ice cartridges and releases it gradually around the insulated milk chamber without continuous external power.

**WHAT IS NEW**

- Modular cartridges: capacity set per load
- Ice inside the milk: direct, fast heat transfer
- Easy to clean: smooth, wide-mouth liner

**Performance**

|                                                                      |        |        |     |
|----------------------------------------------------------------------|--------|--------|-----|
| 30–40 L                                                              | 4–8 °C | 6–12 h | 0 W |
| PS requirements: insulated below 5 °C for 12 hrs, best 14 h at 35 °C |        |        |     |

**Optional monitoring:** Probe → ESP8266 → LED / phone

SIH Idea Submission - Template

Slide 2 – Proposed solution

0:12 – 0:30

## The core idea

### SHOW

- Point at the **ICE / PCM** cartridge in the diagram, then follow the blue arrows.
- Point at the red arrows, "heat leak slowed by PUF".
- End on the four boxes: 30–40 L · 4–8 °C · 6–12 h · 0 W.

### SAY

"Our solution is MilkShield: a double-walled, foam-insulated can with a food-grade liner. Reusable ice cartridges hang from the lid, right inside the milk – the cold goes straight in, and the foam keeps the heat out. No compressor, no electricity."

**Why it works:** "ice inside the milk" is the one idea that sets you apart, so say it slowly.

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**TECHNICAL APPROACH**

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**Workflow:** MILKING → FILL → INSERT COOLING → SEAL → PASSIVE COOLING → TEMP CHECK → TRANSPORT → VILLAGE COLLECTION CENTRE

**THERMAL SIMULATION** (40 L, 35 °C day)

**CARTRIDGES NEEDED** (2 L each)

|                        |   |
|------------------------|---|
| Pre-chilled milk: 12 h | 2 |
| Fresh 30 L: 6 h        | 6 |
| Fresh 40 L: 12 h       | 8 |

**Notes:** Fresh milk addition is 1% of the load, so cartridge count sets capacity. Made by: robotmilk shell + liner → PUF pour → assemble.

SIH Idea Submission - Template

Slide 3 – Technical approach

0:30 – 0:45

## How a farmer uses it

### SHOW

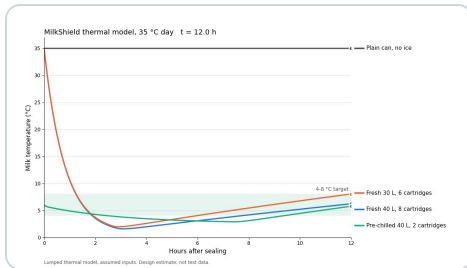
- Run the cursor along the top flow: Milking → Fill → Insert cooling → Seal → ... → Collection centre.
- Finish on the **Cartridges needed** table (2 / 6 / 8).

### SAY

"Fill, drop in frozen cartridges, seal, and go. At the collection centre, cartridges are swapped for frozen ones. The cartridge count is our design knob: two for pre-chilled milk, six for 30 litres, eight for 40."

**Why it works:** it shows you have thought about the real daily routine, not just the gadget.

# Scene by scene – the simulation proof



Clip: sim1\_lumped\_animation.mp4 (12 s)

0:45 – 1:00

## Proof 1 – Thermal model

### SHOW

- Play the clip: the lines draw themselves over 12 hours.
- Grey line (plain can) stays at 35 °C; the **blue line** (40 L, 8 cartridges) drops into the green 4–8 °C band.
- Add a small corner caption: **"Simulation"**.

### SAY

"We sized it through simulation at three levels. First, a thermal model: on a 35 °C day, a plain can stays at 35. MilkShield brings 40 litres below 8 °C in 75 minutes and holds it there for over 14 hours."

**Why it works:** a moving before/after comparison is the fastest way to show performance.

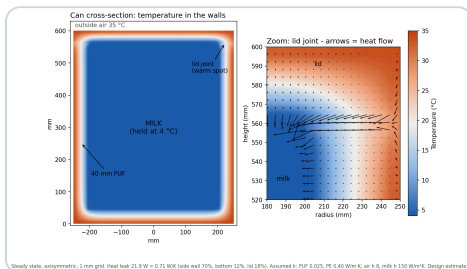


Image: sim3\_wall\_heatmap.png

1:00 – 1:12

## Proof 2 – Heat flow through the walls

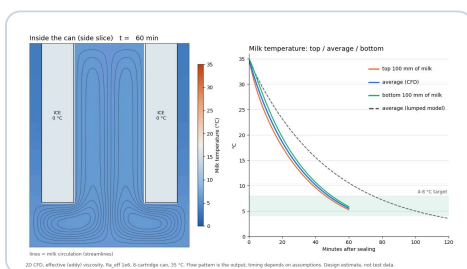
### SHOW

- Show the full image for 2 s, then slowly zoom into the **right panel** (lid joint with arrows).
- Blue = cold milk, orange = hot outside air; the foam holds the whole change.

### SAY

"Second, a heat-flow model of the walls. The foam carries the full temperature drop, and the model found the one weak spot – the lid joint – which our design insulates."

**Why it works:** finding and fixing a weakness shows engineering maturity, and judges reward that.



Clip: sim4\_cfd\_convection.mp4 (12 s), then sim4\_cfd\_snapshots.png

1:12 – 1:30

## Proof 3 – CFD of the milk

### SHOW

- Play the clip: the can turns from orange to blue.
- Trace the flow lines with the cursor: **down the ice** → **along the bottom** → **up the centre**.
- Point at the right-hand graph, where the top, average and bottom curves stay together.
- Cut to the 4-panel snapshot for 3 s.

### SAY

"Third, CFD of the milk itself. Cold milk slides down the ice, sweeps the bottom and rises through the centre – the can stirs itself. Top and bottom stay within about two and a half degrees, and the CFD cools even faster than our thermal model, so our numbers are on the safe side."

**Why it works:** CFD is the visual judges expect from a serious hardware team, and "stirs itself" is memorable.

# Scene by scene – feasibility, impact and close

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FEASIBILITY AND VIABILITY



| Analysis of the feasibility of the idea                       | Potential challenges and risks                  | Strategies for overcoming these challenges                 |
|---------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------|
| <b>Cooling duration</b><br>Simulated: 14 h below 8 °C         | Fresh milk needs >3h the cooling of a 12 h hold | Cartridge count sized per load by simulation (2-8)         |
| <b>Heat leakage</b><br>+12 °F at 35 °C (wall heat-flow model) | Lid joint is the main weak spot (>11 % of risk) | Insulated lid, gasket, no metal through-paths              |
| <b>Weight</b><br>Manageable                                   | <18 kg of ice for a full 40 L, 12 h trip        | Carry only what the load needs; swap at collection centre  |
| <b>Hygiene</b><br>Food-grade HDPE, easy to clean              | Milk residue, little cleaning water             | Smooth liner, wide mouth, crown fits out                   |
| <b>Leakage - rough roads</b><br>Robust                        | Bumps, tipping, cartridge or lid leaks          | Rotomoulded PE, gasket + latches, welded cartridges        |
| <b>Cost - manufacturing</b><br>Visible                        | Tooling cost at volume                          | Prototype from off-the-shelf drums, rotomould + PU-F laser |

TECHNICALLY FEASIBLE  
Simulation + prior art (1)

ECONOMICALLY FEASIBLE  
Prototype + 10-13 h (est.)

OPERATIONALLY FEASIBLE  
Cartridges swapped at centre

MANUFACTURABLE  
Rotomould + PU-F process

SCALABLE  
Cartridge count sets capacity

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Slide 4 – Feasibility and viability

1:30 – 1:42

## It is practical

SHOW

- Point at the first row, "Simulated: 14 h below 8 °C".
- Sweep across the five green boxes at the bottom.

SAY

"It's practical: food-grade materials, a prototype for 6 to 13 thousand rupees, and rotomoulding for scale."

**Why it works:** it answers "can this actually be built and afforded?" before anyone asks.

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IMPACT AND BENEFITS



**TARGET USERS**

- Small & marginal dairy farmers
- Village milk collection centres
- Dairy cooperatives
- Rural milk transporters
- Remote hilly / NE dairy networks



Photo: A person is shown using the milk collection system, which is a small, portable container that can be used in remote areas.

**BENEFITS**

|                                                                                                                                     |                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SOCIAL</b> <ul style="list-style-type: none"><li>Safe milk without electricity</li><li>Suits hilly &amp; NE villages</li></ul>   | <b>ECONOMIC</b> <ul style="list-style-type: none"><li>Reusable icePCM, no power bill</li><li>Fewer rejected milk lots</li></ul>             |
| <b>OPERATIONAL</b> <ul style="list-style-type: none"><li>Easy to clean, moulded handles</li><li>Cartridges sized per trip</li></ul> | <b>ENVIRONMENTAL</b> <ul style="list-style-type: none"><li>No refrigerant, no diesel</li><li>Reusable cartridges, recyclable HDPE</li></ul> |

**MILK JOURNEY**

Scale: later with the same design: 10-40L → 50-100L collection → cooperative bulk

FARMER → MILKSHIELD (cartridge in, sealed) → RURAL TRANSPORT (sealed tank, 5-15 °C) → COLLECTION CENTRE (check temp, unseal) → DAIRY COLD CHAIN (20°C cooling, chilling)

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Slide 5 – Impact and benefits → then Slide 6

1:42 – 1:55

## Who benefits, and why we win

SHOW

- Slide 5: point at the photo of today's uninsulated cans, then the milk-journey strip.
- Cut to Slide 6 and point at the **MilkShield column**, especially "Modular: Yes".

SAY

"It serves small farmers, collection centres and remote hilly and North-East villages – safe milk with no power bill and no diesel. And it's the only modular design among existing options."

**Why it works:** it links the tech to real people, then ends on a clear difference from competitors.

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RESEARCH AND REFERENCES



|                 | Conventional can (1) | Ice-water prototype (1)                         | PCM pad, ICE-PCM (2, 3, 4)          | MilkShield (prototype)                              |
|-----------------|----------------------|-------------------------------------------------|-------------------------------------|-----------------------------------------------------|
| Cooling         | None                 | Ice compartment                                 | None (PCM) passive                  | IcePCM cartridges in milk                           |
| Power           | None                 | None in use                                     | None in use                         | None in use                                         |
| Reported result | No cooling           | 20 L, +17 °C in +90 RH% (20 L, +13 °C for 12 h) | 37 → +10 °C in 30-40 min (reported) | Simulated: below 8 °C in 75 min; held 14 h at 35 °C |
| Modular         | No                   | No                                              | No                                  | Yes: 2-8 cartridges                                 |
| Cost            | + ₹2,200 (40 L)      | Not reported                                    | + ₹1,800-2,000                      | Prototype + ₹6-13 k (est.)                          |

**REFERENCES**

[1] Kumar Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[2] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[3] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[4] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[5] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[6] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[7] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[8] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[9] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

[10] Prasad et al. Int. J. Refrigeration 99 (2019) 20-31. [doi:10.1016/j.sci.2019.04.001](#)

Simulation study: thermal model - wall heat-flow model - 2D CFD (milk cools evenly, top to bottom within 2-3 °C). Next: prototype field trial.

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Slide 6 – hold on the green "Simulation study" box

1:55 – 2:05

## Next step and close

SHOW

- Slow zoom onto the green **Simulation study** box.
- Optional last frame: team name + "Thank you".

SAY

"Next, we build the prototype for field trials at a village collection centre. Thank you."

**Why it works:** ending on a concrete next step sounds confident and ready to execute.

# Explaining the simulations in plain words

Use these if you want to say a little more, or if a judge asks how the simulations work.

## 1 · THERMAL MODEL – "HOW COLD, HOW FAST, HOW LONG"

Treats the milk and the ice cartridges as two linked heat stores. Heat leaks in through the walls, and the ice soaks it up while melting at 0 °C. Solved minute by minute over 24 hours, it gives the temperature curve and the number of cartridges for each load.

**Key results:** below 8 °C in 75 min · held 14 h+ at 35 °C · still 10 h+ on a 45 °C day · 8 cartridges for 40 L, 6 for 30 L, 2 for pre-chilled milk.

**One line to say:** "It's the energy budget of the can, solved minute by minute."

## 2 · WALL HEAT-FLOW MODEL – "WHERE THE HEAT GETS IN"

A 1 mm-resolution heat-conduction model of the can's cross-section: outer shell, foam, liner, lid and lid joint.

**Key results:** about 22 W leaks in at 35 °C · roughly 70 % through the side wall, 18 % through the lid, 12 % through the bottom · the solid-plastic lid joint alone is about 11 %, so we insulate it and keep metal out of the wall.

**One line to say:** "It shows exactly where heat sneaks in, so we fix that spot in the design."

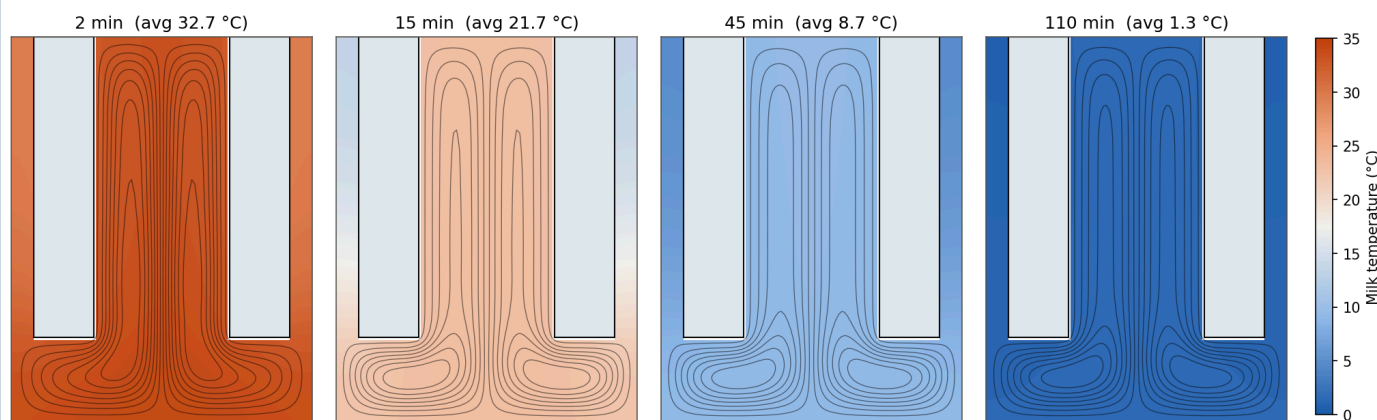
## 3 · CFD – "DOES ALL THE MILK GET COLD, OR ONLY THE MILK NEAR THE ICE?"

A 2D natural-convection simulation of a vertical slice through the can. Cold milk is heavier, so it sinks along the cartridges and sets up a circulation loop without any moving parts.

**Key results:** the whole can cools together · top and bottom within about 2.5 °C · below 8 °C in about 48 min (faster than the thermal model, so our 75 min is the cautious figure) · the last spot to cool is the centre-bottom, so cartridges should reach close to the floor.

**One line to say:** "The ice makes the milk stir itself, so the whole can cools evenly."

Cold milk runs down the ice, sweeps the floor and rises through the centre: a self-stirring loop (2D CFD)



## Likely judge questions – ready answers

|                                        |                                                                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Which software did you use?"          | "Our own Python models: a thermal model, a finite-volume heat-conduction model of the walls, and a 2D CFD solver. Every assumption is listed at the top of each file, so every number can be traced."                                                                  |
| "How reliable are the results?"        | "Three separate models point the same way: the CFD cools even faster than the thermal model, and the wall model's heat leak is lower than the value we designed for. So the figures on our slides are on the cautious side. The prototype trials will fine-tune them." |
| "Have you built it?"                   | "The design is sized and checked by simulation. The prototype is our next phase, starting from off-the-shelf food-grade drums, then field trials at a collection centre."                                                                                              |
| "Won't the milk go below 4 °C?"        | "It settles around 2 °C, because ice sits at 0 °C. That's safe: Codex sets only an upper limit, and milk freezes at about -0.5 °C, so it cannot freeze."                                                                                                               |
| "What about a 45 °C day?"              | "The same 8 cartridges still hold the milk below 8 °C for over 10 hours in our simulation."                                                                                                                                                                            |
| "How heavy is it?"                     | "A full 40-litre trip needs 8 cartridges – 16 kg of ice. Smaller loads carry fewer."                                                                                                                                                                                   |
| "Why not ANSYS?"                       | "At the idea stage, a transparent model we can explain line by line is more useful. With the prototype we'll move to a full 3D model to refine the lid and cartridge layout."                                                                                          |
| "Why is it better than existing cans?" | "Plain cans don't cool at all; existing prototypes use one fixed ice or PCM jacket. MilkShield puts the ice inside the milk and is modular – the farmer carries only what the load needs."                                                                             |

### SAY IT LIKE THIS

- "Our simulation shows..." / "Simulated at 35 °C..."
- "The design is sized by three thermal models."
- "Our figures are on the cautious side."
- "The prototype is our next phase."

### AVOID

- "We tested it" or "it achieved" – these imply physical tests.
- Reading the slides word for word.
- Rushing the CFD clip – let the colour change play out.
- Talking over the clips with numbers not shown on screen.

### RECORDING CHECKLIST

- ☐ Record slides in PowerPoint slideshow mode (full screen, 16:9).
- ☐ Insert **sim1\_lumped\_animation.mp4** and **sim4\_cfd\_convection.mp4** at 0:45 and 1:12.
- ☐ Keep each still on screen at least 5 s; use a slow zoom on the heat map and CFD snapshots.
- ☐ Add a small "Simulation" caption in the corner of the three simulation shots.
- ☐ Use a quiet room and an earphone mic; record voice separately if possible.
- ☐ Rehearse once with a timer. If it runs long, shorten the 1:42 impact line first.
- ☐ Export at 1080p or 720p MP4 and watch it once end to end before uploading.